Assessing the Association of Genetics, Cognitive and General Function, and Imaging Biomarkers with Risk of Progression from Mild Cognitive Impairment to Dementia

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0.1 Project Summary

This project analyzes how clinical, genetic, and imaging biomarkers relate to the risk of progression from mild cognitive impairment (MCI) to dementia in Alzheimer's disease. Using data from the Alzheimer's Disease Neuroimaging Initiative (ADNI), I applied survival analysis techniques to estimate how different predictors influence time to disease progression. A Cox proportional hazards model achieved strong predictive performance (concordance 0.88). Cognitive scores—particularly ADAS13—were among the strongest predictors of progression, alongside amyloid status and hippocampal volume. This project demonstrates how survival modeling can be used to integrate multi-domain health data and identify key drivers of disease progression.

0.2 Introduction

Alzheimer's disease is a leading cause of dementia, with progression often beginning at the stage of mild cognitive impairment (MCI). However, the rate at which individuals progress from MCI to dementia varies widely. Understanding which factors are most associated with progression risk can help improve early intervention strategies and risk stratification. In this project, I use ADNI data to evaluate how cognitive scores, genetic factors (APOE4), and brain imaging biomarkers relate to time-to-dementia using survival analysis.

0.3 Methods

0.3.1 Data Overview

The dataset comes from the Alzheimer's Disease Neuroimaging Initiative (ADNI), a longitudinal study tracking individuals with mild cognitive impairment.

Sample size: 604 participants Outcome: Time (years) to progression to dementia Predictors: Demographics: age, gender, education Genetics: APOE4 allele count Cognitive scores: ADAS13, CDRSB, MMSE, ADNI-Mem, ADNI-EF Imaging: amyloid status, hippocampus, ventricles, entorhinal volume

Participants who did not progress to dementia during the study period were treated as right-censored at their last follow-up time.

Because approximately 25% of observations had missing data, the primary analysis used complete-case data. This approach assumes data are missing completely at random (MCAR). Sensitivity analyses using simple imputation were conducted to assess robustness. Results were not significantly changed when using imputed dataset

1 1. Setup

libraries

```
library(tidyr)
library(dplyr)
library(ggplot2)
library(readr)
library(KMsurv)
library(survival)
library(car)
library(knitr)
library(gtsummary)
```

2 2. Load Data

```
data <- read_csv("./data/jack_msexam_data.csv")
```

3 3. Clean + Prepare Data

```
# add a categorical age covariate because of non-linear relationship with outcome
data_clean <- data %>%
  mutate(
    amyloid = factor(AMYLOID_STATUS, labels = c("Amyloid_negative","Amyloid_positive")),
    gender  = factor(PTGENDER, labels = c("Female","Male")),
    dementia = factor(Dementia, labels = c("MCI","Dementia")),
    alleles = factor(APOE4, labels = c("Zero","One","Two")),
    age_categorical = cut(AGE, c(0,60,65,70,75,80,100))
  ) %>%
  select(-AMYLOID_STATUS, -PTGENDER, -APOE4)

colnames(data_clean)[c(3,13)] <- c('Education', 'D_status')
```

missing handling

```
# Preserve full dataset before dropping missing
data_full <- data_clean
data_clean <- drop_na(data_clean)
```

4 4. Feature Engineering

standardization

```
standardize <- function(x) {
  (x - mean(x, na.rm = TRUE)) / sd(x, na.rm = TRUE)
}
# Create standardized version of data to compare of effect size more intuitively
data_standardized <- data_clean %>%
  mutate(across(c(AGE, Education, CDRSB_bl, ADAS13_bl, MMSE_bl,
                  Ventricles_bl, Hippocampus_bl, Entorhinal_bl),
                standardize,
                .names = "{.col}_standardized"))
```

5 5. Exploratory Analysis

Table 1

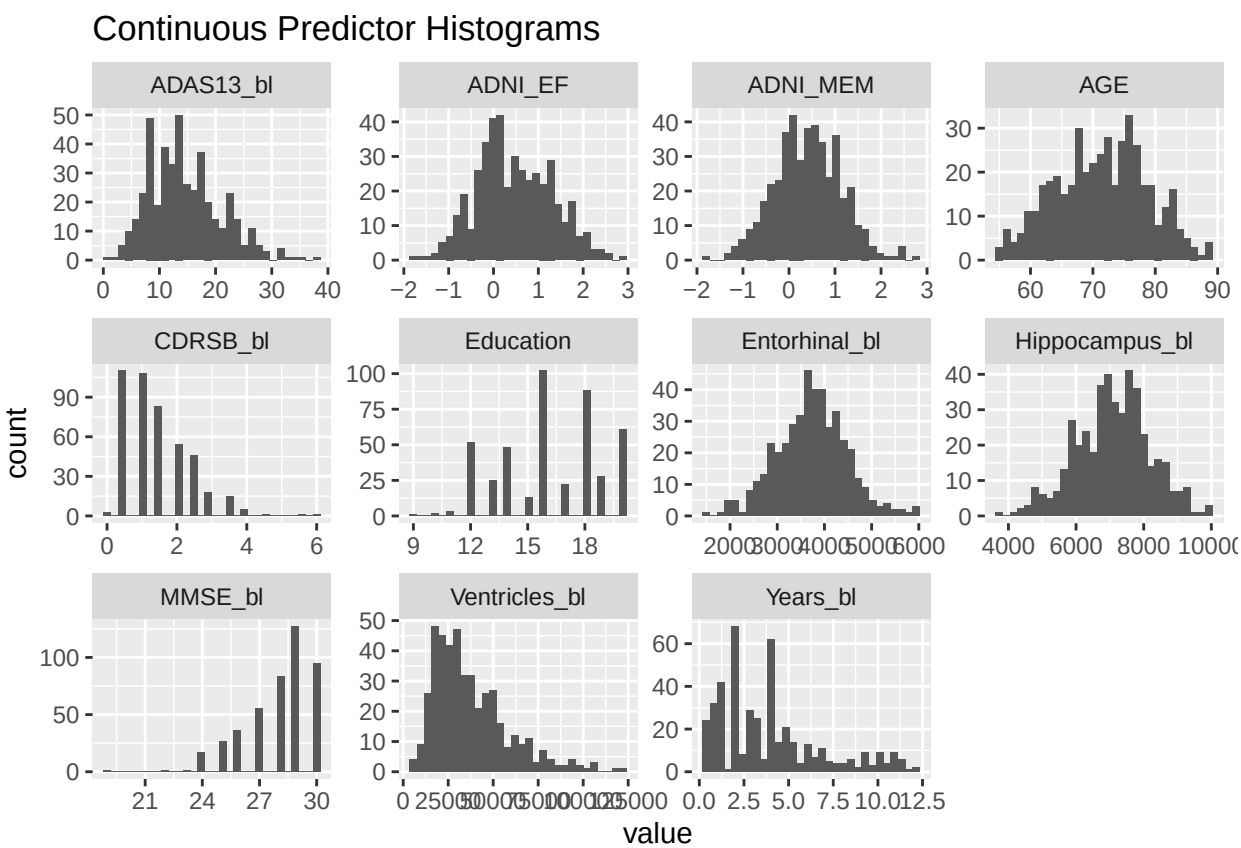
```r
# Compare the whole data set to the subset of complete data
data_clean$dataset <- "Complete Cases"
data_full$dataset <- "All Cases"

combined_data <- bind_rows(data_clean, data_full)
Table1<- combined_data %>%
  select(
    AGE, age_categorical, Education, gender, Years_bl, dementia, amyloid, alleles,
    CDRSB_bl, ADAS13_bl, MMSE_bl,
    Ventricles_bl, Hippocampus_bl, Entorhinal_bl, ADNI_EF, ADNI_MEM,
    dataset
  ) %>%
  tbl_summary(by = dataset,
    statistic = list(
      all_continuous() ~ "{mean} ({sd})",
      all_categorical() ~ "{n} ({p}%)"
    ),
    missing = "no"
  ) %>%
  bold_labels()
```

```r
Table1 %>% as_kable()
```

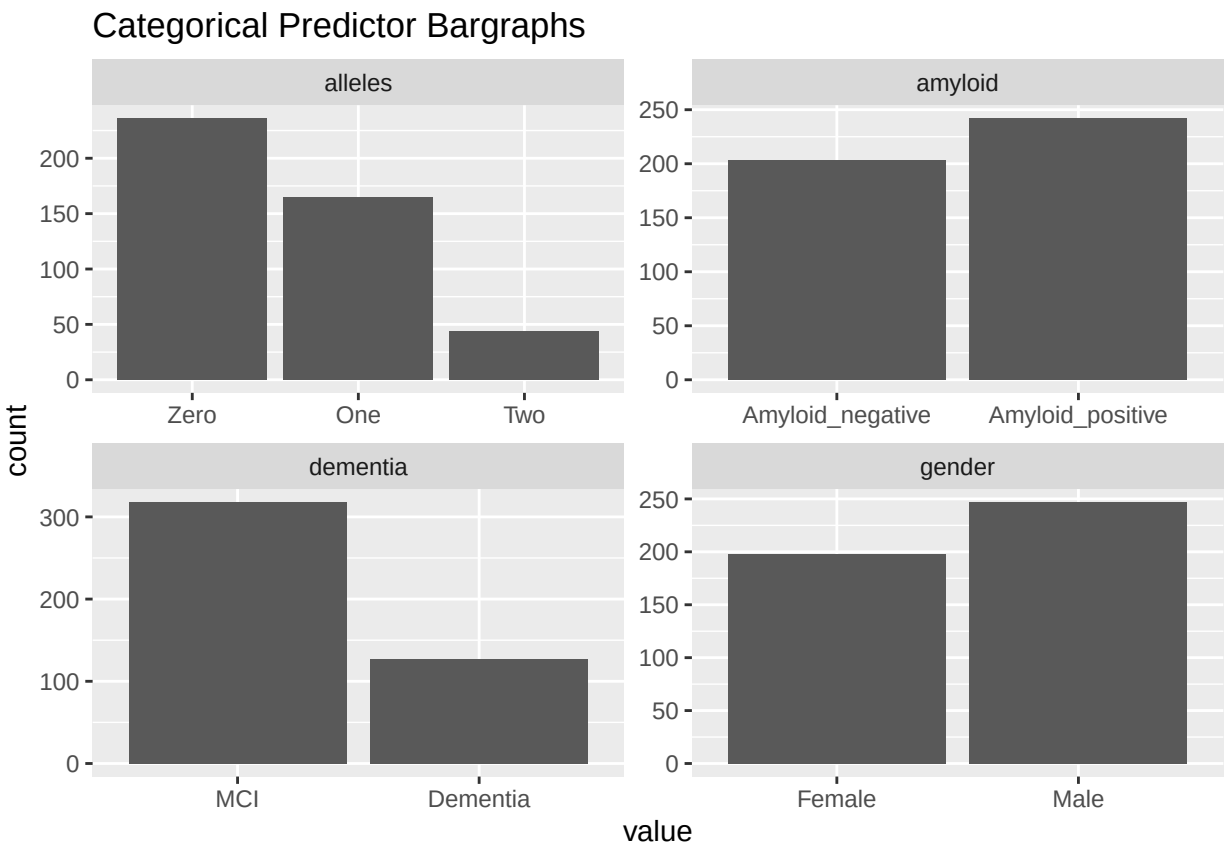
Characteristic	All Cases N = 604	Complete Cases N = 445
AGE	72 (7)	71 (7)
age_categorical		
(0,60]	32 (5.3%)	27 (6.1%)
(60,65]	90 (15%)	71 (16%)
(65,70]	119 (20%)	92 (21%)
(70,75]	141 (23%)	101 (23%)
(75,80]	134 (22%)	99 (22%)
(80,100]	85 (14%)	55 (12%)
Education	16.24 (2.58)	16.22 (2.62)
gender		
Female	262 (43%)	198 (44%)
Male	342 (57%)	247 (56%)
Years_bl	3.80 (2.89)	3.95 (2.87)
dementia		
MCI	438 (73%)	318 (71%)
Dementia	166 (27%)	127 (29%)
amyloid		
Amyloid_negative	278 (46%)	203 (46%)
Amyloid_positive	320 (54%)	242 (54%)
alleles		
Zero	295 (52%)	236 (53%)
One	211 (37%)	165 (37%)
Two	58 (10%)	44 (9.9%)
CDRSB_bl	1.48 (0.95)	1.47 (0.92)
ADAS13_bl	15 (7)	15 (6)
MMSE_bl	27.97 (1.77)	28.04 (1.75)
Ventricles_bl	38,206 (21,431)	37,202 (20,688)
Hippocampus_bl	7,090 (1,099)	7,101 (1,090)
Entorhinal_bl	3,737 (757)	3,731 (735)
ADNI_EF	0.40 (0.85)	0.46 (0.84)
ADNI_MEM	0.37 (0.74)	0.41 (0.72)

5.1 distributions

```
data_clean %>%
  pivot_longer(col = c(2:12), names_to = 'var' , values_to = 'value'  ) %>%
  ggplot(mapping = aes(x = value)) +
  geom_histogram() +
  facet_wrap(~var, , scales = 'free')+
  labs(title = "Continuous Predictor Histograms")
```




```
data_clean %>%
  pivot_longer(col = c(14:17), names_to = 'var' , values_to = 'value'  ) %>%
  ggplot(mapping = aes(x = value)) +
  geom_bar() +
  facet_wrap(~var, scales = 'free')+
  labs(title = "Categorical Predictor Bargraphs")
```



APOE4 allele count and amyloid status were strongly correlated, so they were not included together in the same models to avoid redundancy. Other predictors showed low multicollinearity (VIF < 2), suggesting they could be included together without instability.

```
knitr::kable(
  table(data_full$alleles, data_full$amyloid),
  caption = 'Alleles and Plaque Distribution'
)
```

Table 2: Alleles and Plaque Distribution

	Amyloid_negative	Amyloid_positive
Zero	196	95
One	54	157
Two	7	50

```
chisq.test(table(data_full$alleles, data_full$amyloid))
```

```
##
##  Pearson's Chi-squared test
##
## data:  table(data_full$alleles, data_full$amyloid)
## X-squared = 114.9, df = 2, p-value < 2.2e-16
```

6 6. Modeling

final Cox model

6.0.1 Methods of Analysis

I used survival analysis to model time to progression from MCI to dementia. Model: Cox proportional hazards regression Baseline model: demographic variables (age, education, gender) Full model: added cognitive, genetic, and imaging predictors Model selection: based on AIC and concordance Diagnostics: proportional hazards and residual checks Kaplan-Meier curves were used to compare survival across key groups and check key model assumptions (amyloid status, APOE4 alleles).

```
# Create survival object for dementia time/status
ADNI.surv <- with(data_standardized, Surv(Years_bl,D_status))
```

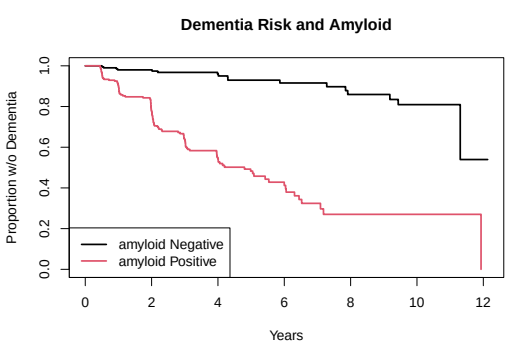
6.1 Non-parametric model checks

6.1.1 create object

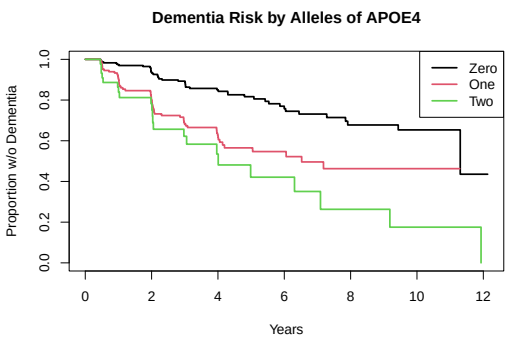
```
# fit non parametiric survival models
amyloidsurv <- survfit(ADNI.surv~amyloid, data=data_standardized)
allelesurv <- survfit(ADNI.surv~alleles, data=data_standardized)
```

6.1.2 Non parametic Plots

```
# Check proportional hazards assumption of cox model
plot(amyloidsurv,col = 1:2,lwd=2)
title("Dementia Risk and Amyloid", xlab = 'Years', ylab = 'Proportion w/o Dementia')
legend("bottomleft", c('amyloid Negative','amyloid Positive') ,col=1:2,lwd=2)
```




```
plot(survfit(ADNI.surv~alleles,data=data_standardized),col = 1:3,lwd=2)
title("Dementia Risk by Alleles of APOE4", xlab = 'Years', ylab = 'Proportion w/o Dement
legend("topright",c("Zero","One","Two"),col=1:3,lwd=2)
```



6.2 Initial fit

```
cox_base <- coxph(ADNI.surv~  gender + AGE + Education, data = data_clean)

summary(cox_base)
```

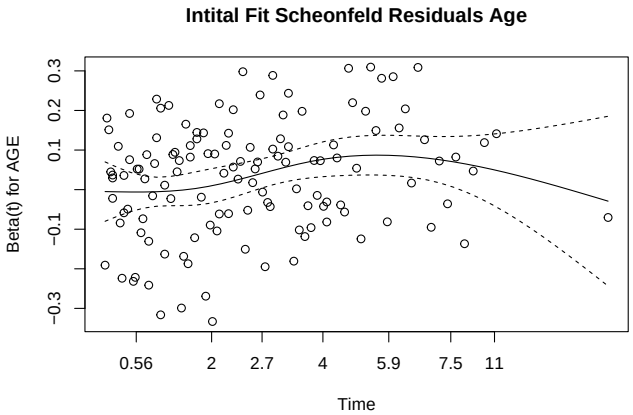
```
## Call:
## coxph(formula = ADNI.surv ~ gender + AGE + Education, data = data_clean)
##
##   n= 445, number of events= 127
##
##                  coef  exp(coef)   se(coef)      z Pr(>|z|)
## genderMale  0.0006445  1.0006447  0.1844982  0.003   0.9972
## AGE         0.0313271  1.0318229  0.0126780  2.471   0.0135 *
## Education  -0.0489656  0.9522139  0.0346240 -1.414   0.1573
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##            exp(coef) exp(-coef) lower .95 upper .95
## genderMale    1.0006     0.9994    0.6970     1.437
## AGE           1.0318     0.9692    1.0065     1.058
## Education     0.9522     1.0502    0.8897     1.019
##
## Concordance= 0.56  (se = 0.029 )
## Likelihood ratio test= 8.41  on 3 df,   p=0.04
## Wald test            = 8.49  on 3 df,   p=0.04
## Score (logrank) test = 8.51  on 3 df,   p=0.04
```

```
drop1(cox_base, test="Chisq")
```

```
## Single term deletions
##
## Model:
## ADNI.surv ~ gender + AGE + Education
##           Df    AIC    LRT Pr(>Chi)
## <none>       1379.8
```

```
## gender      1 1377.8 0.0000  0.99721
## AGE         1 1384.0 6.1033  0.01349 *
## Education   1 1379.8 1.9825  0.15912
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Check the need for stratification using Schoenfeld residuals
base_Schoenfeld <- cox.zph(cox_base)

plot(base_Schoenfeld[2])
title(main = 'Intital Fit Scheonfeld Residuals Age')
```




```
cox_stratified_base <- coxph(ADNI.surv~Education + strata(age_categorical), data = data_
zph2<- cox.zph(cox_stratified_base)
```

6.2.1 initial fit check

Martingale residuals suggest the remaining base covariates are the acceptable as linear predictors of hazard.

6.3 full model

```
cox_full <- update(cox_stratified_base, ADNI.surv~Education +Ventricles_bl+Hippocampus_b

cox_reduced <- update(cox_stratified_base, ADNI.surv~Education +Hippocampus_bl+amyloid+
```

```
# Check for collinearity in the covariates
vif_table <- as.data.frame(vif(cox_full))
kable(vif_table, caption = "Variance Inflation Factors")
```

Table 3: Variance Inflation Factors

	GVIF	Df	GVIF^(1/(2*Df))
Education	1.129720	1	1.062883

	GVIF	Df	GVIF^(1/(2*Df))
Ventricles_bl	1.194230	1	1.092808
Hippocampus_bl	1.695424	1	1.302085
Entorhinal_bl	1.604305	1	1.266612
amyloid	1.191773	1	1.091684
ADAS13_bl	2.665792	1	1.632725
CDRSB_bl	1.140261	1	1.067830
MMSE_bl	1.410840	1	1.187788
ADNI_EF	1.202719	1	1.096685
ADNI_MEM	3.281316	1	1.811440
alleles	1.345641	2	1.077041
strata(age_categorical)	12.020461	0	Inf

```r
## Check covariate significance to arrive at reduced model
# drop1 full model
drop1_full <- as.data.frame(drop1(cox_full, test = "Chisq"))
kable(drop1_full, caption = "Drop1 Results - Full Model")
```

Table 4: Drop1 Results - Full Model

	Df	AIC	LRT	Pr(>Chi)
	NA	699.6925	NA	NA
Education	1	697.9847	0.2921592	0.5888402
Ventricles_bl	1	698.9643	1.2717845	0.2594322
Hippocampus_bl	1	701.8815	4.1890010	0.0406871
Entorhinal_bl	1	699.5756	1.8830903	0.1699833
amyloid	1	734.7721	37.0795623	0.0000000
ADAS13_bl	1	707.9786	10.2861341	0.0013403
CDRSB_bl	1	728.7600	31.0675077	0.0000000
MMSE_bl	1	700.9148	3.2222682	0.0726429
ADNI_EF	1	707.0865	9.3939966	0.0021770
ADNI_MEM	1	699.0226	1.3301037	0.2487868
alleles	2	700.9888	5.2963415	0.0707806
strata(age_categorical)	0	1116.0793	416.3867885	NA

```r
# anova comparison
anova_table <- as.data.frame(anova(cox_reduced, cox_full, test = "Chisq"))
kable(anova_table, caption = "Model Comparison")
```

Table 5: Model Comparison

loglik	Chisq	Df	Pr(>|Chi|)
-343.6356	NA	NA	NA
-337.8462	11.57865	6	0.0720565

```r
# drop1 reduced model
drop1_reduced <- as.data.frame(drop1(cox_reduced, test = "Chisq"))
kable(drop1_reduced, caption = "Drop1 Results - Reduced Model")
```

Table 6: Drop1 Results - Reduced Model

	Df	AIC	LRT	Pr(>Chi)
	NA	699.2711	NA	NA
Education	1	698.1018	0.8306325	0.3620897
Hippocampus_bl	1	713.8773	16.6061180	0.0000460
amyloid	1	754.8572	57.5860118	0.0000000
ADAS13_bl	1	731.9420	34.6708274	0.0000000
CDRSB_bl	1	731.0451	33.7739634	0.0000000
ADNI_EF	1	707.4521	10.1809680	0.0014190
strata(age_categorical)	0	1120.6052	421.3340695	NA

Characteristic	HR	95% CI	p-value
Education	1.03	0.96, 1.11	0.4
Hippocampus_bl_standardized	0.61	0.49, 0.78	<0.001
amyloid			
Amyloid_negative	—	—	
Amyloid_positive	6.71	3.76, 12.0	<0.001
ADAS13_bl_standardized	1.95	1.56, 2.44	<0.001
CDRSB_bl_standardized	1.67	1.42, 1.97	<0.001
ADNI_EF	0.64	0.49, 0.85	0.002

Abbreviations: CI = Confidence Interval, HR = Hazard Ratio

7 7. Results

model summary

```
cox_reduced_standardized <- update(cox_reduced,
                                   ADNI.surv~Education +Hippocampus_bl_standardized+amyloid+
                                   data= data_standardized)

tbl_regression(cox_reduced_standardized, exponentiate = TRUE)
```

plots

7.1 Results

Key Findings Amyloid-positive individuals had significantly higher risk of progression Risk increased with number of APOE4 alleles Cognitive scores were among the strongest predictors: ADAS13 performed best Followed by ADNI-Mem and CDRSB Imaging biomarkers (especially hippocampal volume) were also strong predictors

Final Model The final Cox model included: Amyloid status Hippocampal volume ADAS13, CDRSB, and ADNI-EF scores Education (with age stratification) This model achieved: Concordance: 0.89 Interpretation: Amyloid positivity increased risk 6-7x Higher ADAS13 and CDRSB scores increased risk Higher ADNI-EF and hippocampal volume were protective

7.2 Discussion:

Discussion This analysis highlights the value of combining information from cognitive, imaging, and biomarker tests to assess risk of dementia in individuals Alzheimers. Although APOE4 is a known genetic risk factor, its predictive value was reduced when amyloid status was included, likely due to strong correlation between the two. The final model suggests that a relatively small set of features can capture much of the variation in progression risk, which has practical implications for designing efficient screening or monitoring strategies.

7.2.1 Limitations

Missing data reduced the usable sample size Results are based on ADNI participants and may not generalize to broader populations Potential bias from study dropout cannot be ruled out

8 References

Chapleau, M., Iaccarino, L., Soleimani-Meigooni, D., & Rabinovici, G. D. (2022). The role of amyloid PET in imaging Neurodegenerative Disorders: a review. Journal of Nuclear Medicine, 63(Supplement 1), 13S-19S. https://doi.org/10.2967/jnumed.121.263195

Crane, P. K., Narasimhalu, K., Gibbons, L. E., Mungas, D. M.,Haneuse, S., Larson, E. B., et al. (2008). Item response theoryfacilitated cocalibrating cognitive tests and reduced bias in estimated rates of decline. Journal of Clinical Epidemiology, 61(10), 1018–1027 e1019

Crane, P.K., Carle, A, Gibbons, L. E., Insel, P., Mackin, R. S., Gross, A., Jones, R. N., Mukherjee, A., Curtis, S. M., Harvey, D., Weiner, M., Mungas, D. (2012). Development and assessment of a composite score for memory in the Alzheimer's Disease Neuroimaging Initiative (ADNI). Brain Imaging and Behavior (2012) 6:502–516.

Geriatric nursing resources. (n.d.). HIGN. https://hign.org/consultgeri/resources (https://www.psychdb.com/_media/mmse.pdf)

Gibbons, L. E., Carle, A. C. , Mackin, R. S. , Harvey, D., Mukherjee, S., Insel, P., Curtis S. M., Mungas, D., Crane, P. K. (2012). A composite score for executive functioning, validated in Alzheimer's Disease Neuroimaging Initiative (ADNI) participants with baseline mild cognitive impairment. Brain Imaging and Behavior (2012) 6:517–527

Lu X, Zhang Y, Tang Y, Bernick C, Shan G. Conversion to Alzheimer's disease dementia from normal cognition directly or with the intermediate mild cognitive impairment stage. Alzheimer's Dement. 2025; 21:e14393. https://doi.org/10.1002/alz.14393

New Facts & Figures Report on Mild Cognitive Impairment | Mayo Clinic Connect. (n.d.). Mayo Clinic Connect. https://connect.mayoclinic.org/blog/living-with-mild-cognitive-impairment-mci/newsfeed-post/new-facts-figures-report-on-mild-cognitive-impairment/

Nyunt, Ma Shwe Zin; Chong, Mei Sian; Lim, Wee Shiong; Lee, Tih Shih; Yap, Philip; Ng, Tze Pin (2013). "Reliability and Validity of the Clinical Dementia Rating for Community-Living Elderly Subjects without an Informant". Dementia and Geriatric Cognitive Disorders Extra. 3 (1): 407–416.

Weiner M.W., Kanoria S., Miller M.J., Aisen P.S., Beckett L.A., Conti C., Diaz A., Flenniken D., Green R.C., Harvey D.J., Jack C.R., Jagust W., Lee E.B., Morris J.C., Nho K., Nosheny R., Okonkwo O.C., Perrin R.J., Petersen R.C., Rivera-Mindt M., Saykin A.J., Shaw L.M., Toga A.W.,Tosun D.,Veitch D.P. (2024). Overview of Alzheimer's Disease Neuroimaging Initiative and future clinical trials.The Journal of Altzheimer's Association (2024) 10.1002/alz.14321